

Fabián A. Campos Montero

Geotechnical Engineer & Researcher — Infrastructure, Soil Mechanics & AI

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Profile

Fabián is a geotechnical engineer and researcher at Deltares with more than 10 years of experience across geotechnical engineering, site investigation, monitoring, natural hazards and applied research. His work combines a strong geotechnical foundation with AI and data methods, from CPT based subsurface schematisation and machine learning to field investigation, laboratory testing and fibre optic monitoring. He also works on slope and embankment stability, probabilistic landslide assessment and numerical modelling. Across research and consultancy projects, he develops practical methods that turn complex geotechnical data into clearer engineering decisions.

Experience

Geotechnical Researcher and Consultant

December 2022 – Present

Deltares

Delft, The Netherlands

- Helped strengthen the technical basis for assessing Dutch railway embankments in ProRail's RESET programme by combining site investigation, laboratory and centrifuge data with PLAXIS 2D/3D, limit equilibrium analyses and Python automation, supporting decisions on safely accommodating more, faster and heavier trains.
- Developed machine learning methods that relate dark fibre measurements along railway tracks to free field accelerometer data, creating a scalable way to screen larger parts of the network for locations with higher vibration risk.
- Advanced automated subsurface interpretation by developing SchemaGAN, a generative AI method that creates geotechnical cross sections from limited CPT data. Related work includes GeoSyn for synthetic ground data and Deep Reinforcement Learning for site investigation planning.
- Made site characterisation and numerical studies faster and more reproducible by building Python workflows around CPT data, statistical parameterisation, PLAXIS and further developing the DGEOLIB Plus Deltares software.
- Turned complex geotechnical and data analyses into clear engineering results for clients, researchers and multidisciplinary teams.

Geotechnical Engineer, Researcher and Lecturer

2014 – 2021

University of Costa Rica - Geomechanics Laboratory

San José, Costa Rica

- Supported land use and risk decisions through probabilistic landslide studies based on PISA methods, combining field, laboratory and GIS data to assess instability under dry, saturated and seismic scenarios at municipal, community and site scales.
- Planned and carried out site investigations using CPT, SPT, boreholes, sampling, geological mapping, MASW, seismic refraction and electrical resistivity tomography, combining the results with laboratory testing to build ground models for foundations, slopes and infrastructure.
- Assessed natural slopes, landfills, mines, quarries and infrastructure sites by developing geotechnical models and carrying out static, pseudostatic and limit equilibrium analyses.
- Supported safe and practical foundation design by interpreting field and laboratory data to assess bearing capacity, settlement and expected ground behaviour for residential, institutional and infrastructure projects.
- Built the soil parameter basis for design and stability work by performing and interpreting laboratory testing, including direct shear, triaxial and oedometer tests, across volcanic, residual and alluvial soils.
- Lectured in soil mechanics and geology, supervised student research and mentored junior staff in field investigation, laboratory work, GIS and geotechnical analysis.

Geotechnical Risk Consultant*Inter-American Development Bank**2018 – 2020**South America*

- Supported remediation planning for the Kantutani landslide in La Paz by bringing experience in municipal solid waste mechanics, helping define terms of reference for follow up investigations, coordinating with the municipality and developing geotechnical risk recommendations.
- Reduced uncertainty around Quito Metro spoil management after a fatal slope failure by helping define terms of reference for further geotechnical investigation, assessing the affected disposal site and reviewing safer alternatives.
- Mapped landslides along Route RP33 in Salta through field surveys and GIS, creating a corridor inventory of instability mechanisms and priority risk areas.
- Turned the RP33 hazard assessment into practical risk management actions by defining terms of reference for further studies and contributing recommendations for slope monitoring, early warning and short term risk reduction.

Geotechnical Engineer Intern*Royal HaskoningDHV**August – November 2022**The Netherlands*

- Compared conventional and load distributed anchor systems for the Oosterweel Link project using analytical calculations and PLAXIS, helping assess how anchor length affects performance and the use of double grouted anchors.

Technical Support Specialist*GeoStru Software**2018 – 2019**Part time*

- Improved software adoption for geotechnical users in Latin America by producing technical training material, supporting users, testing new features and giving structured feedback to the development team.

Intern Geologist*Diquís Hydroelectric Project, ICE**2013**Costa Rica*

- Supported geological characterisation of the planned reservoir area through geological and geomorphological mapping, in situ permeability testing, data interpretation and GIS database development.

Education**Delft University of Technology***MSc in Geo-Engineering / Geotechnical Engineering**2023*

Thesis on Deep Learning for Geotechnical Engineering: The Effectiveness of Generative Adversarial Networks in Subsoil Schematization.

International Geotechnical Centre*Diploma in Underground and Overground Geomechanics**2020***University of Costa Rica***MSc in Engineering Geology**2017*

Thesis on the geomechanical characterisation and slope stability of a soil and municipal solid waste system at the San Ramón landfill.

University of Costa Rica*BSc in Geology**2015***Distinctions**

- **Schreuders Study Prize 2023 – Honorable Mention** for the MSc thesis on deep learning for geotechnical subsurface schematisation.
- **Cum laude**, Diploma in Underground and Overground Geomechanics, International Geotechnical Centre, 2020.
- **Honorable Mention**, MSc in Engineering Geology thesis, University of Costa Rica, 2017.

Selected Publications

- Linnemann, Y. S., Campos Montero, F., Zuada Coelho, B., van Duinen, A., & Fern, J. (2026). Assessment of critical train speeds of railways on soft soils – a case study. *E3S Web of Conferences*, 730, 05001. DOI: [10.1051/e3sconf/202673005001](https://doi.org/10.1051/e3sconf/202673005001) [↗](#).
- Post, M., Campos Montero, F., Schweckendiek, T., de Jong, E., & Kleinjan, A. (2026). Application of line loads versus block loads for railway embankment stability assessment [Accepted conference paper, 24–26 August 2026]. *International Conference on Advances and Innovations in Soft Soil Engineering*.
- Campos Montero, F. A., Zuada Coelho, B., Smyrniou, E., Taormina, R., & Vardon, P. J. (2025). Schemagan: A conditional generative adversarial network for geotechnical subsurface schematisation. *Computers and Geotechnics*, 183, 107177. DOI: [10.1016/j.compgeo.2025.107177](https://doi.org/10.1016/j.compgeo.2025.107177) [↗](#).
- Campos Montero, F., Smyrniou, E., Zuada Coelho, B., Taormina, R., & Vardon, P. J. (2025). GEOSYN: Synthetic Geotechnical Cross-Sections for Machine Learning Applications. *29th European Young Geotechnical Engineers Conference*. DOI: [10.32762/eygec.2025.23](https://doi.org/10.32762/eygec.2025.23) [↗](#).
- Zuada Coelho, B., Smyrniou, E., & Campos Montero, F. A. (2025). Optimizing geotechnical in-situ site investigations using Deep Reinforcement Learning. *Geodata and AI*, 100038. DOI: [10.1016/j.geoai.2025.100038](https://doi.org/10.1016/j.geoai.2025.100038) [↗](#).
- Campos, F., Mora, R., Sequeira, J., & Garnier, M. (2021). Application of kinematic admissibility analysis and limit equilibrium to optimize the excavation geometry of a quarry in the canton of León Cortés, San José, Costa Rica [Article in Spanish]. *Revista Geológica de América Central*, 1–25. DOI: [10.15517/rgac.v66i0.49786](https://doi.org/10.15517/rgac.v66i0.49786) [↗](#).

Technical Expertise

Site Investigation & Geodata: CPT, SPT, boreholes, soil and rock sampling, geological and geomorphological mapping, MASW, seismic refraction, electrical resistivity tomography, LiDAR, photogrammetry, GIS and ground modelling

Laboratory & Soil Mechanics: Direct shear, triaxial testing, oedometer and consolidation, index testing, soil classification, strength and compressibility interpretation

Monitoring & Instrumentation: Distributed fibre optic sensing, DAS, accelerometers, inclinometers, pore water pressure monitoring, railway vibration monitoring, event detection, signal processing and monitoring system validation

AI & Data Methods: Python, machine learning, generative AI, reinforcement learning, synthetic geotechnical data, subsurface schematisation, uncertainty and spatial variability, automated data processing and statistical analysis

Numerical & Analytical Geotechnics: PLAXIS 2D/3D, Python API automation, limit equilibrium methods, probabilistic and stochastic analysis, Monte Carlo simulation, sensitivity and parametric studies, slope and embankment stability, foundation design and back analysis

Software & Reproducibility: Jupyter, Git, version control, Docker, DGEOLIB Plus, Rocscience tools, AutoCAD, LaTeX, collaborative software development, automated reporting and post processing

Natural Hazards & Risk: Landslide mapping, probabilistic slope instability, PISA methods, geotechnical risk assessment, monitoring and early warning concepts, static and seismic slope stability

Languages

Spanish (native) English (fluent, TOEFL iBT 115) Dutch (B1)